

Samuel Clouse

Bioinformatics & Computational Biology | Microbial Genomics | Sustainable Agriculture

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SUMMARY

Bioinformatician and biologist with **5+ years of experience** leveraging **genomic pipelines, data analytics, and automation** to accelerate microbial R&D and sustainable agriculture. Skilled at integrating **DNA sequencing, image analysis, and spatial datasets** to deliver decision-support tools. Experienced collaborating with interdisciplinary teams across **France, the U.S., and the U.K.** to build scalable solutions for **biobanking, strain identification, and crop health**. Passionate about applying computational tools to real-world biological challenges.

CORE SKILLS

Bioinformatics & Genomics: BLAST pipelines, qPCR assay design, sequence analysis

Data Science: Python, Pandas, NumPy, Matplotlib, dashboarding (Plotly / Dash)

Automation & Tools: Image-based spore counting, Excel/Python integration, GUI apps

Geospatial Analysis: Applied use of **ArcGIS & QGIS**

Collaboration: Cross-functional R&D, technical presentations, client engagement

Languages: Fluent English & French

EXPERIENCE

Mycophyto – Grasse, France

Research Engineer | Jan 2023 – Present

- Developed and deployed a **custom bioinformatics pipeline** with a **GUI-based desktop app** to automate **strain identification** using **NCBI BLAST** — cutting identification time **from 3 hours to under 15 minutes per batch**.
- Designed a **qPCR assay** and an **automated spore-counting tool**, increasing throughput **by 35%** and reducing lab costs **by 23%**.
- **Co-invented a patented method** for isolating **native mycorrhizal fungi**, **tripling the in-house biobank**.
- Led **field trials** across vineyards, orchards, and turf systems to test **microbial soil inoculants**, managing **design, logistics, data, and client engagement**.
- Presented findings and strategic insights to **investors, R&D partners, and academic collaborators**.

USDA Agricultural Research Service – Remote

Biological Data Science Intern | Mar 2022 – Aug 2022

- Developed an automated genomics pipeline that **reduced data processing time by over 50%**
- Designed interactive dashboards visualizing trait data across 100+ genotypes to support marker-assisted selection
- Contributed to tools now integrated into legumeinfo.org, improving access to genomic resources.

Iowa State University – Nematology Lab – Ames, IA
Research Assistant II | Jan 2018 – Mar 2021

- Cultured **single-species populations** of **soybean cyst nematodes (SCN)** and isolated **esophageal glands** under an inverted microscope for **RNA extractions**.
- Designed **PCR primers** using **bioinformatics-guided candidate gene selection** derived from **RNA-seq cDNA libraries**.
- Cloned predicted **effector genes** into **E. coli** and **Agrobacterium**, generating **Arabidopsis transgenic lines** to study effector function.
- Performed **yeast two-hybrid assays** to identify **plant protein interaction partners** for effector targets.
- Collaborated closely with bioinformatics teams to **link computational predictions with functional genomics experiments**, bridging **in silico** analyses and **in planta** validation.

Iowa State University – Department of Horticulture – Ames, IA
Research Assistant (Aquaponics Systems) | Jan 2017 – Dec 2017

- Designed, built, and maintained closed-loop aquaponic systems for optimized nutrient flow between tilapia tanks and herb/vegetable beds
- Monitored system data (**nitrate, pH, DO, biomass**) and used analytics to boost production by **15%**.

EDUCATION

MSc – Biocontrol Solutions for Plant Health
Université Côte d'Azur – France | 2021–2023

BSc – Biology
Iowa State University – Ames, IA | 2016–2020
Exchange Year – University of Exeter, UK (2019–2020)

PROJECTS

-  **BioBlaster Tool** | [GitHub](#)
Developed a **desktop app** that parses DNA sequences, automates **NCBI BLAST** queries, and exports annotated results to Excel. Built with **Python**, **Tkinter**, and **Pandas**. Reduced internal strain identification times by **80%**.
-  **Spacebar Counter App** | [GitHub](#)
Created a **Tkinter-based GUI tool** with global hotkeys for counting fungal spores manually. Widely used by lab and remote teams for **high-throughput spore quantification**.

CERTIFICATIONS, AFFILIATIONS & TECHNICAL TRAINING

- **GIS Proficiency: ArcGIS & QGIS**
- **Advanced SCUBA Diver Certification (PADI-equivalent)** — 40m depth
- **SPUN Science Associate** — Global fungal biodiversity initiative